

ABERPORTH BUOY 62301, United Kingdom

WMO: 996850

Lat: 52.300N

Lon: 4.500W

Elev: 0

StdP: 14.70

Time Zone: 0.00 (EUW)

Period: 01-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	35.5	37.2	24.7	18.8	40.2	27.2	21.1	38.8	39.1	48.1	35.7	48.0	15.2	70	0.730

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.1	65.6	62.1	64.1	61.1	63.1	60.4	62.8	64.4	61.9	63.4	61.0	62.7	9.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.9	82.8	63.9	61.0	80.1	63.1	60.1	77.5	62.3	28.3	64.8	27.7	63.6	27.1	62.9	68.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.4	30.2	27.7	DB	33.9	69.0	2.3	2.2	32.3	70.6	30.9	71.9	29.6	73.1	27.9	74.8	(4)
(5)			WB	30.8	65.2	2.1	1.9	29.2	66.5	28.0	67.7	26.8	68.7	25.2	70.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.2	45.6	44.7	45.6	48.2	51.9	56.3	59.3	60.2	59.2	55.8	51.7	47.5	(6)
(7)		DBStd	6.34	3.89	3.52	3.05	2.73	2.48	2.44	2.48	1.75	2.34	3.14	3.40	4.13	(7)
(8)		HDD50	619	139	148	138	67	9	0	0	0	0	3	21	94	(8)
(9)		HDD65	4669	601	567	602	505	406	262	178	149	175	284	398	542	(9)
(10)		CDD50	1428	3	1	1	11	68	188	288	317	276	184	73	18	(10)
(11)		CDD65	2	0	0	0	0	0	0	1	1	0	0	0	0	(11)
(12)		CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	14.5	17.4	15.3	13.9	12.9	13.3	11.6	11.8	12.6	13.7	16.2	17.6	17.9	(14)
(15)	Precipitation	PrecAvg	49.0	5.6	4.0	3.0	3.0	3.0	2.9	3.1	3.6	3.3	5.6	6.1	5.8	(15)
(16)		PrecMax	62.6	9.9	10.1	5.8	6.3	7.1	7.0	7.6	8.2	5.6	10.7	11.3	11.8	(16)
(17)		PrecMin	38.7	0.2	1.1	1.0	0.7	0.4	0.8	0.7	0.2	0.6	1.9	3.1	1.0	(17)
(18)		PrecStd	5.9	2.2	2.0	1.4	1.4	1.4	1.7	1.6	1.9	1.2	2.2	2.2	2.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	52.5	52.8	56.7	61.5	65.5	67.9	67.4	65.9	63.0	58.9	56.7	(19)
(20)			MCWB	51.9	51.1	48.7	52.6	57.9	60.7	63.9	64.1	61.8	59.9	56.5	53.8	(20)
(21)		2%	DB	52.4	51.2	51.0	53.9	58.3	62.9	66.1	64.7	64.4	61.5	58.0	55.5	(21)
(22)			MCWB	50.5	49.7	48.1	50.9	55.1	59.2	62.8	61.8	61.0	58.9	55.6	52.4	(22)
(23)		5%	DB	51.3	50.1	49.9	52.7	56.3	60.7	64.3	63.6	63.2	60.5	57.1	54.3	(23)
(24)			MCWB	49.2	48.4	47.8	50.1	53.5	57.5	61.4	61.0	60.3	58.0	54.8	51.4	(24)
(25)		10%	DB	50.4	49.1	49.2	51.6	55.1	59.4	62.9	62.7	62.3	59.7	56.1	52.9	(25)
(26)			MCWB	48.3	47.2	47.1	49.3	52.7	56.7	60.5	60.5	59.7	57.1	53.5	50.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.6	52.0	49.9	53.0	58.4	61.7	64.5	65.4	62.8	60.5	57.4	54.9	(27)
(28)			MCDWB	53.1	52.4	50.9	55.3	60.4	64.4	67.0	67.2	64.4	61.9	58.1	55.9	(28)
(29)		2%	WB	51.2	50.4	49.0	51.7	55.6	59.5	63.0	62.7	61.8	59.5	56.4	53.8	(29)
(30)			MCDWB	52.1	50.9	50.1	53.4	57.5	61.8	65.2	63.6	63.4	61.0	57.5	55.2	(30)
(31)		5%	WB	50.0	48.7	48.3	50.5	54.3	58.3	61.9	61.8	61.0	58.6	55.2	52.2	(31)
(32)			MCDWB	50.9	49.6	49.5	52.0	55.8	60.2	63.8	63.0	62.7	60.1	56.7	53.8	(32)
(33)		10%	WB	48.9	47.6	47.6	49.7	53.1	57.3	60.7	60.9	60.2	57.7	53.9	50.9	(33)
(34)			MCDWB	50.0	48.8	48.9	51.2	54.5	58.9	62.5	62.4	62.0	59.4	55.7	52.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.2	3.7	3.9	3.8	3.5	3.5	3.1	3.1	3.5	3.9	3.9	4.3	(35)
(36)			MCDBR	4.8	4.1	4.4	5.1	5.6	5.0	4.8	4.0	4.4	4.7	4.6	5.2	(36)
(37)			MCWBR	6.1	4.6	4.5	4.1	4.5	4.4	3.8	3.8	4.1	5.1	5.5	6.7	(37)
(38)		5% WB	MCDBR	4.7	4.0	4.3	4.7	5.2	4.6	4.4	3.8	3.9	4.5	4.5	5.1	(38)
(39)			MCWBR	6.1	4.7	4.6	4.1	4.3	4.4	3.8	4.0	4.2	5.4	5.6	6.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.332	0.344	0.371	0.387	0.394	0.384	0.388	0.371	0.365	0.353	0.330	0.322	(40)	
(41)		taud	2.323	2.375	2.323	2.308	2.338	2.392	2.387	2.448	2.457	2.440	2.381	2.301	(41)	
(42)		Ebn at Noon	206	243	258	269	272	276	272	272	259	237	207	187	(42)	
(43)		Edh at Noon	20	25	32	37	37	36	36	32	28	23	19	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	226	460	853	1350	1681	1825	1649	1383	1008	560	277	167	(44)	
(45)		RadStd	23	42	65	94	111	154	147	91	68	47	24	19	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (52 neighbors)	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A

Nomenclature: See separate page